

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

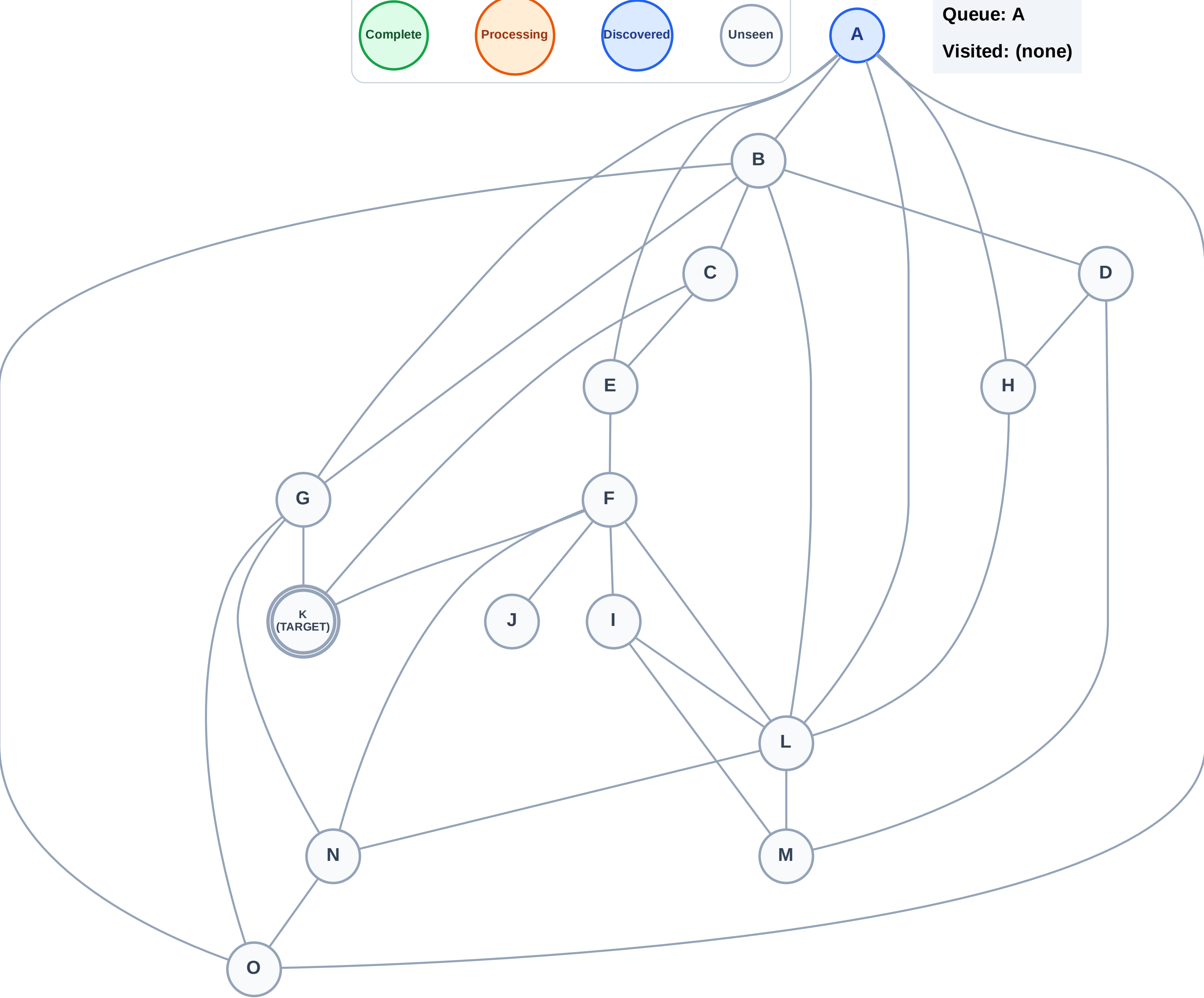
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

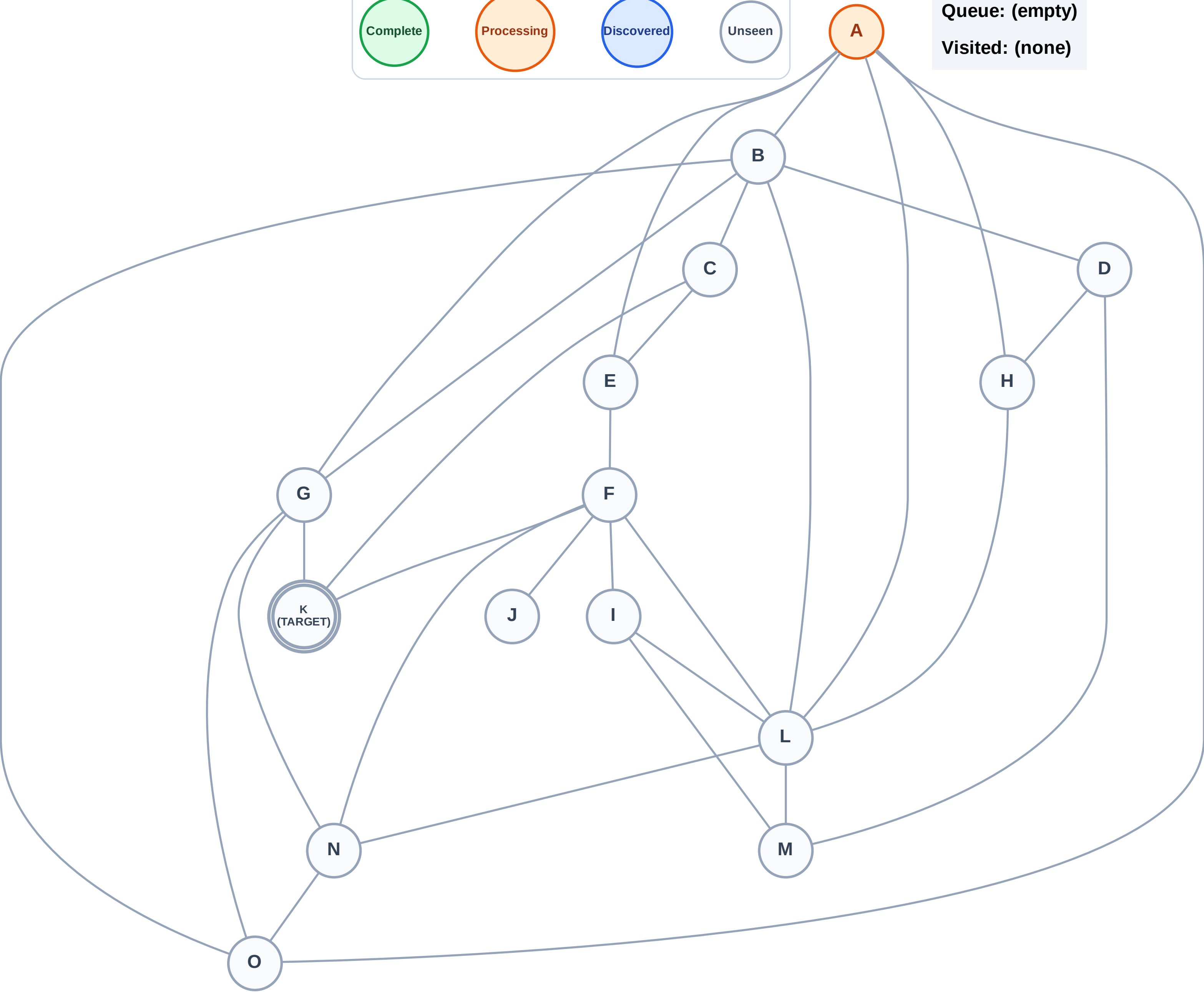
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



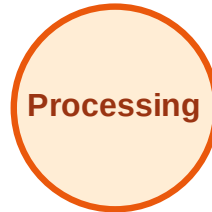
BFS Traversal

Step 3: Discover B, E, G, H, L, O from A

Legend



Complete



Processing



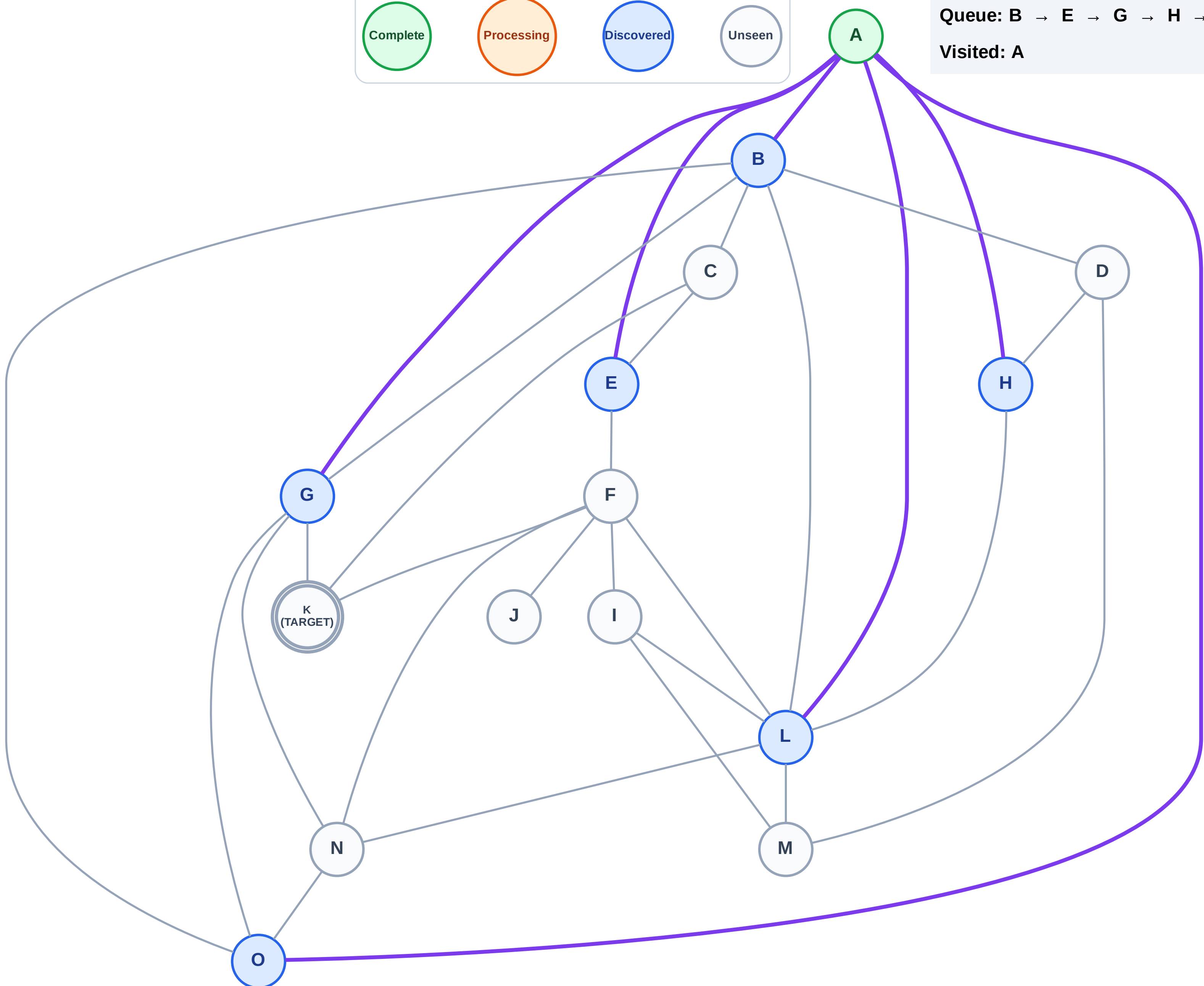
Discovered



Unseen

Queue: B → E → G → H → L → O

Visited: A



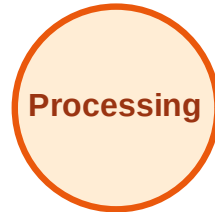
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



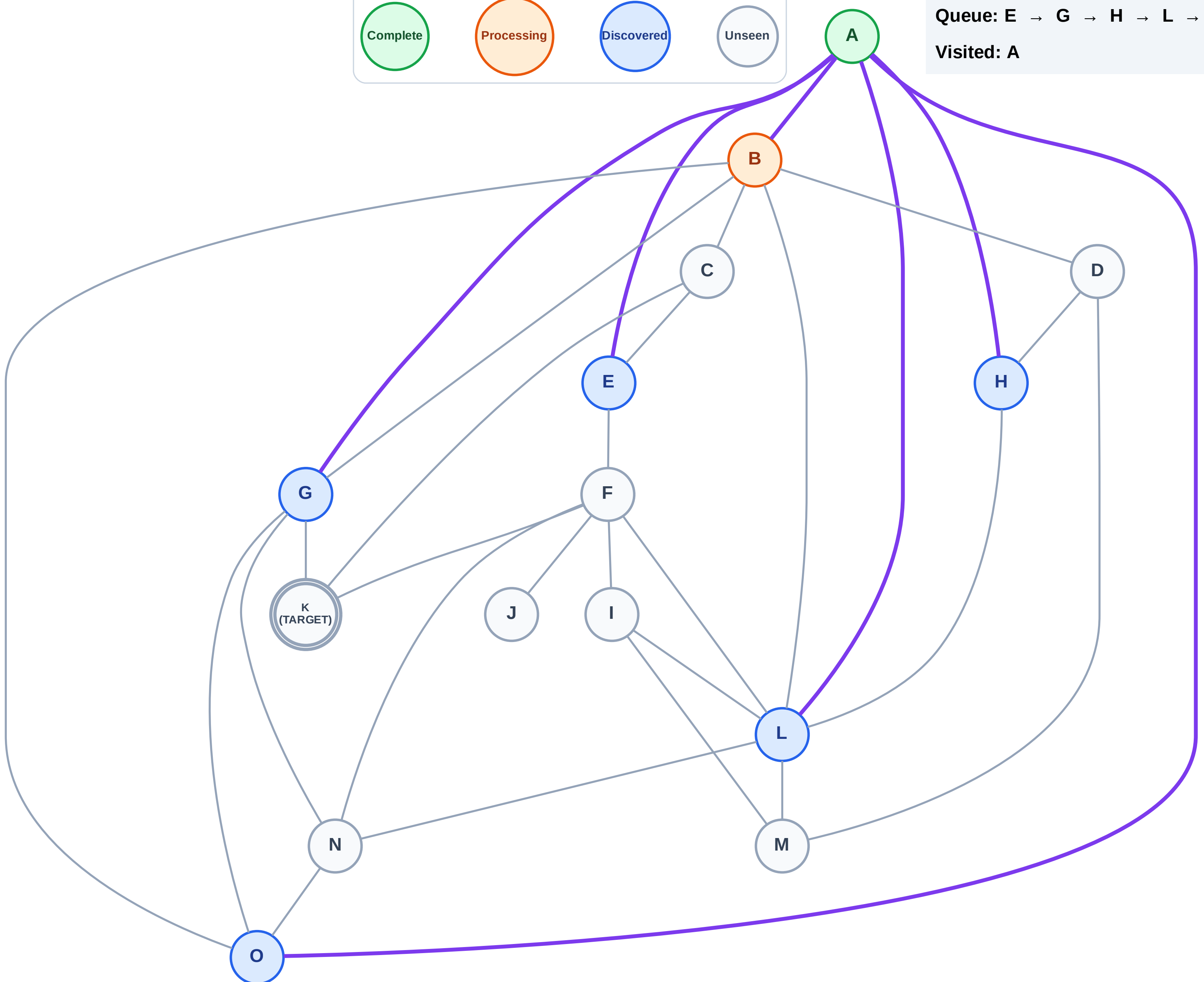
Discovered



Unseen

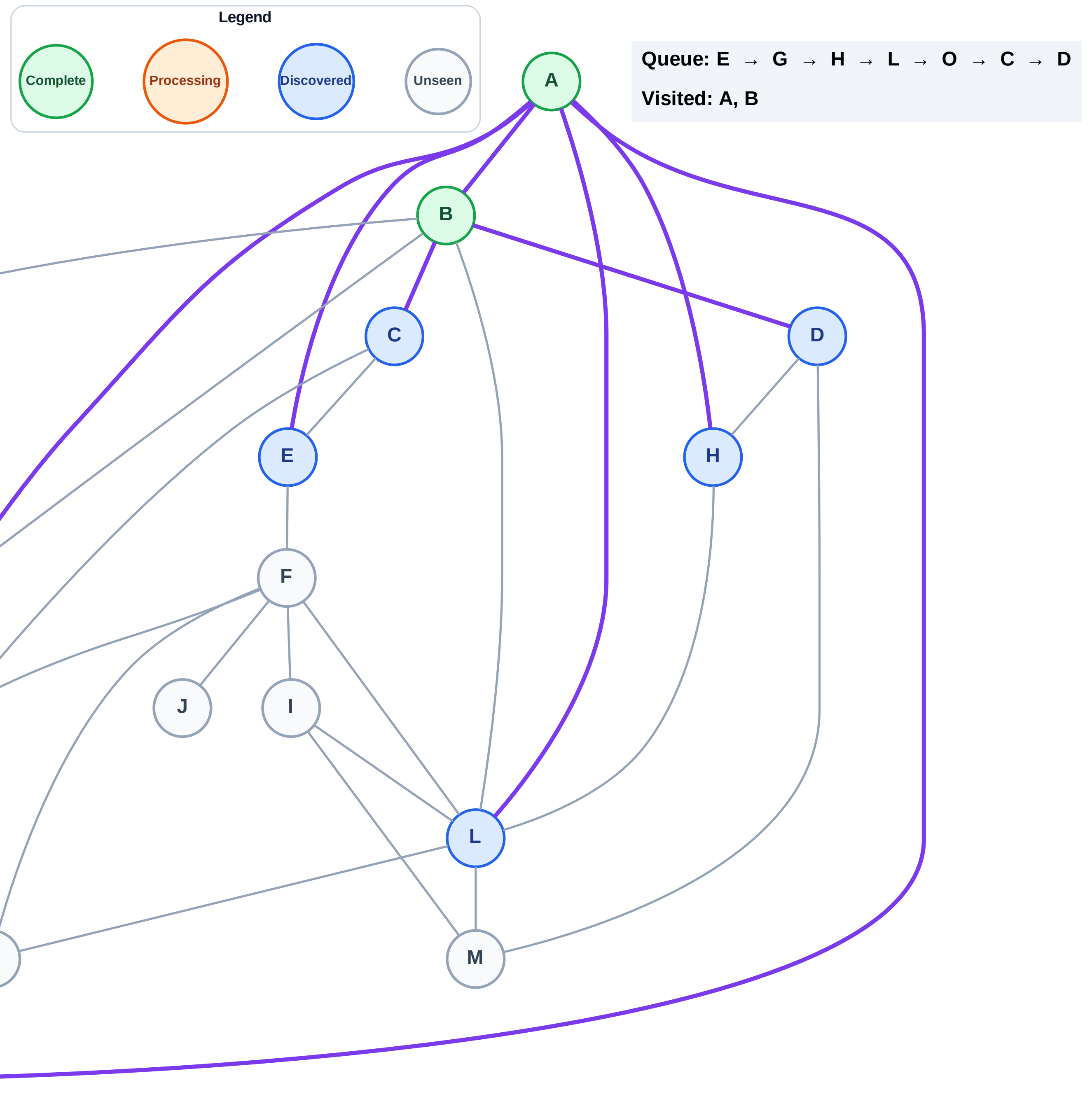
Queue: E → G → H → L → O

Visited: A



Step 5: Discover C, D from B

Step 5: Discover C, D from B



BFS Traversal

Step 6: Process node E

Legend

Complete

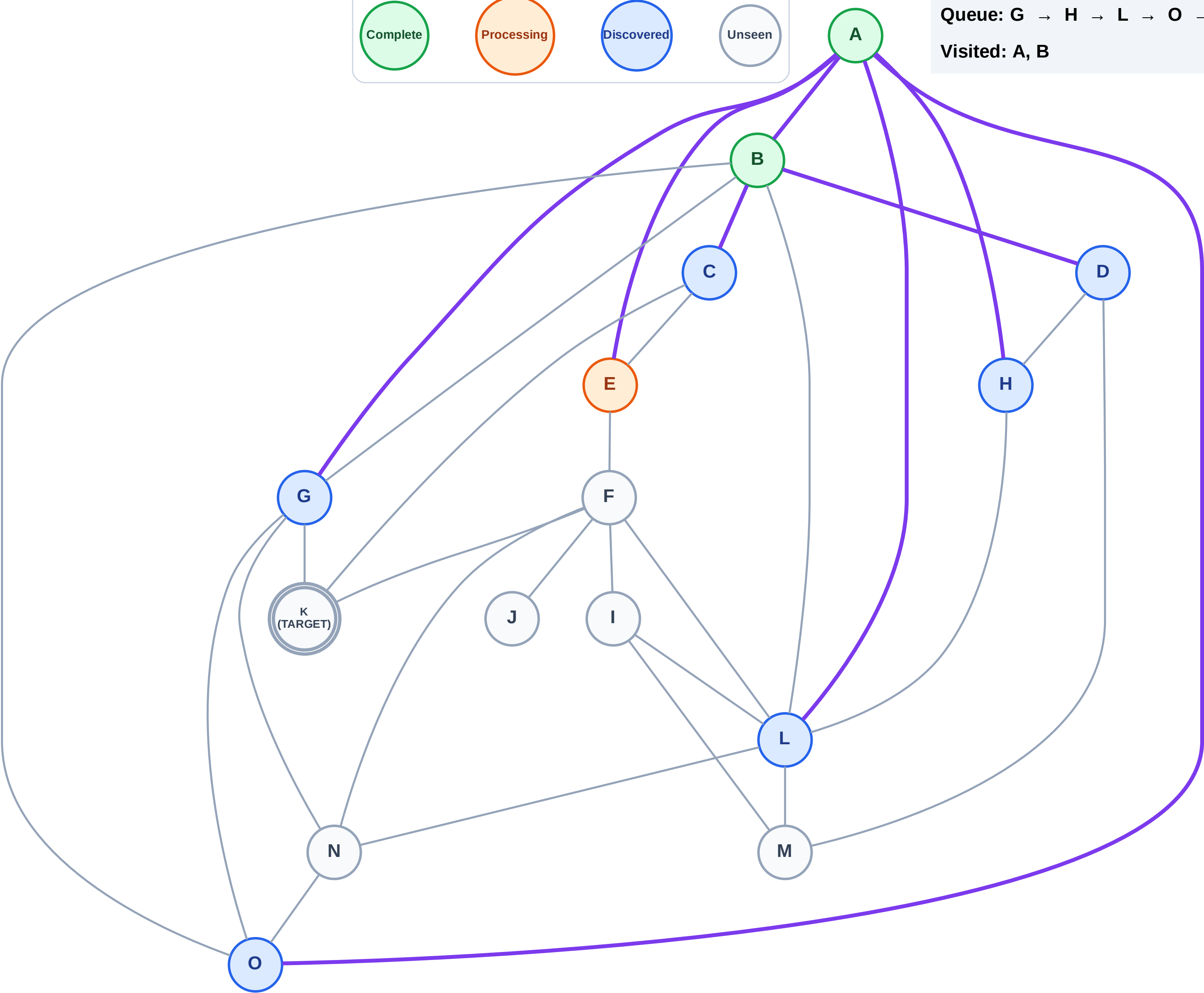
Processing

Discovered

Unseen

Queue: G → H → L → O → C → D

Visited: A, B



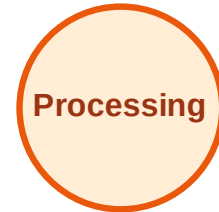
BFS Traversal

Step 7: Discover F from E

Legend



Complete



Processing



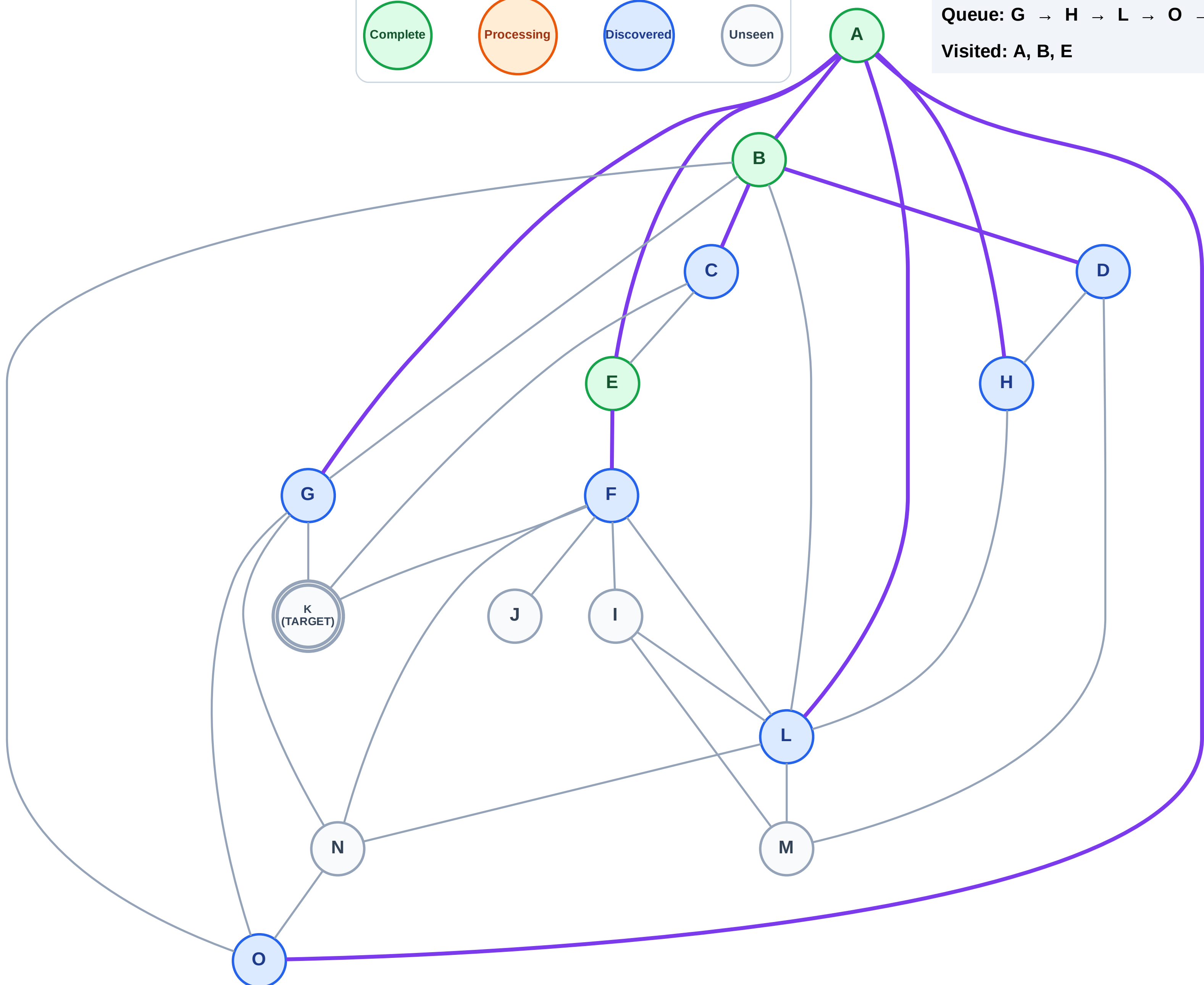
Discovered



Unseen

Queue: G → H → L → O → C → D → F

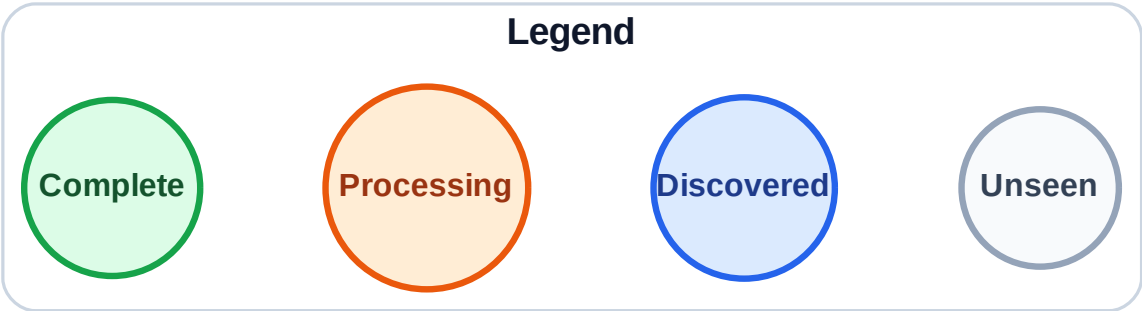
Visited: A, B, E



BFS Traversal

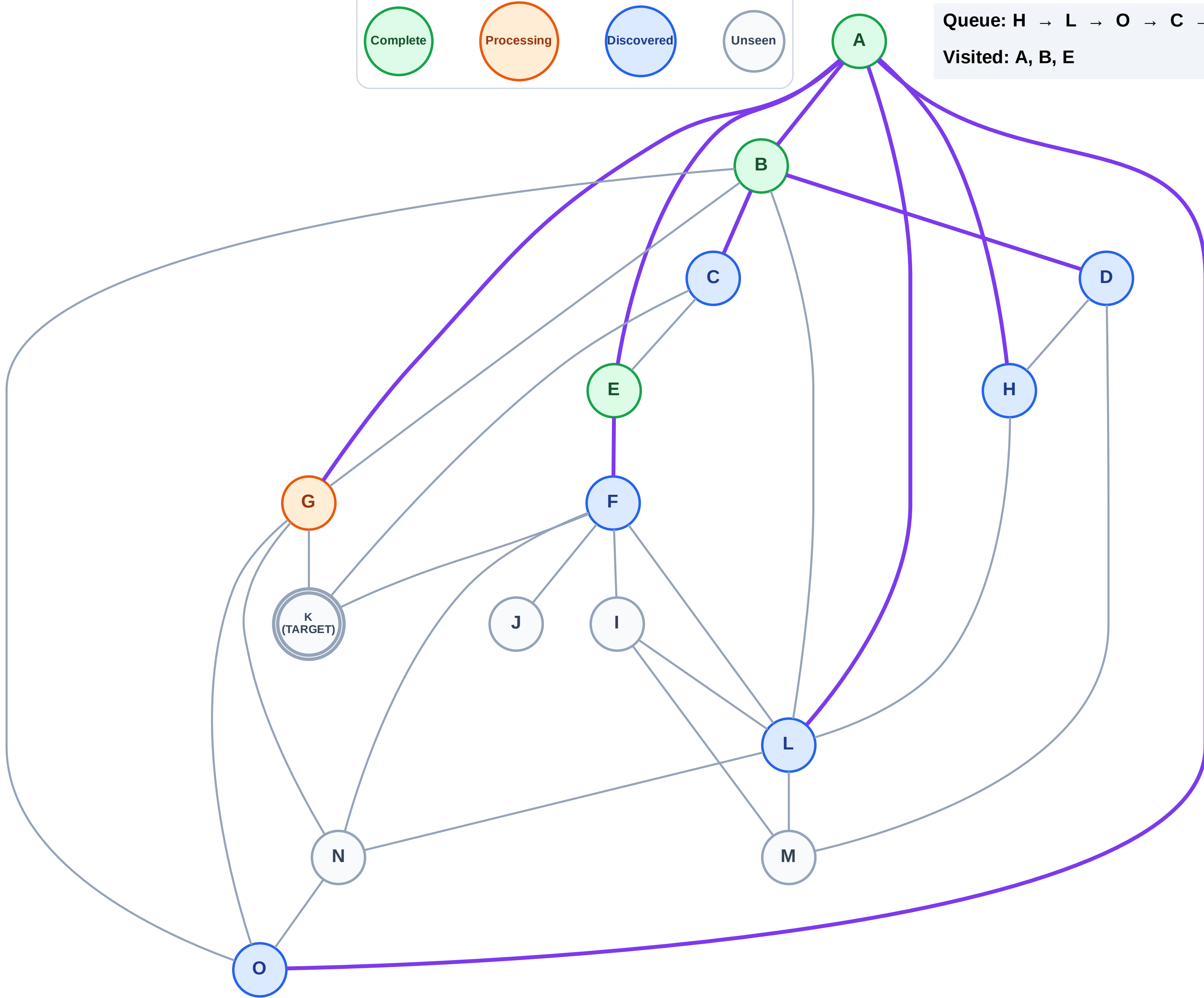
Step 8: Process node G

Legend



Queue: H → L → O → C → D → F

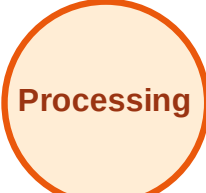
Visited: A, B, E



BFS Traversal

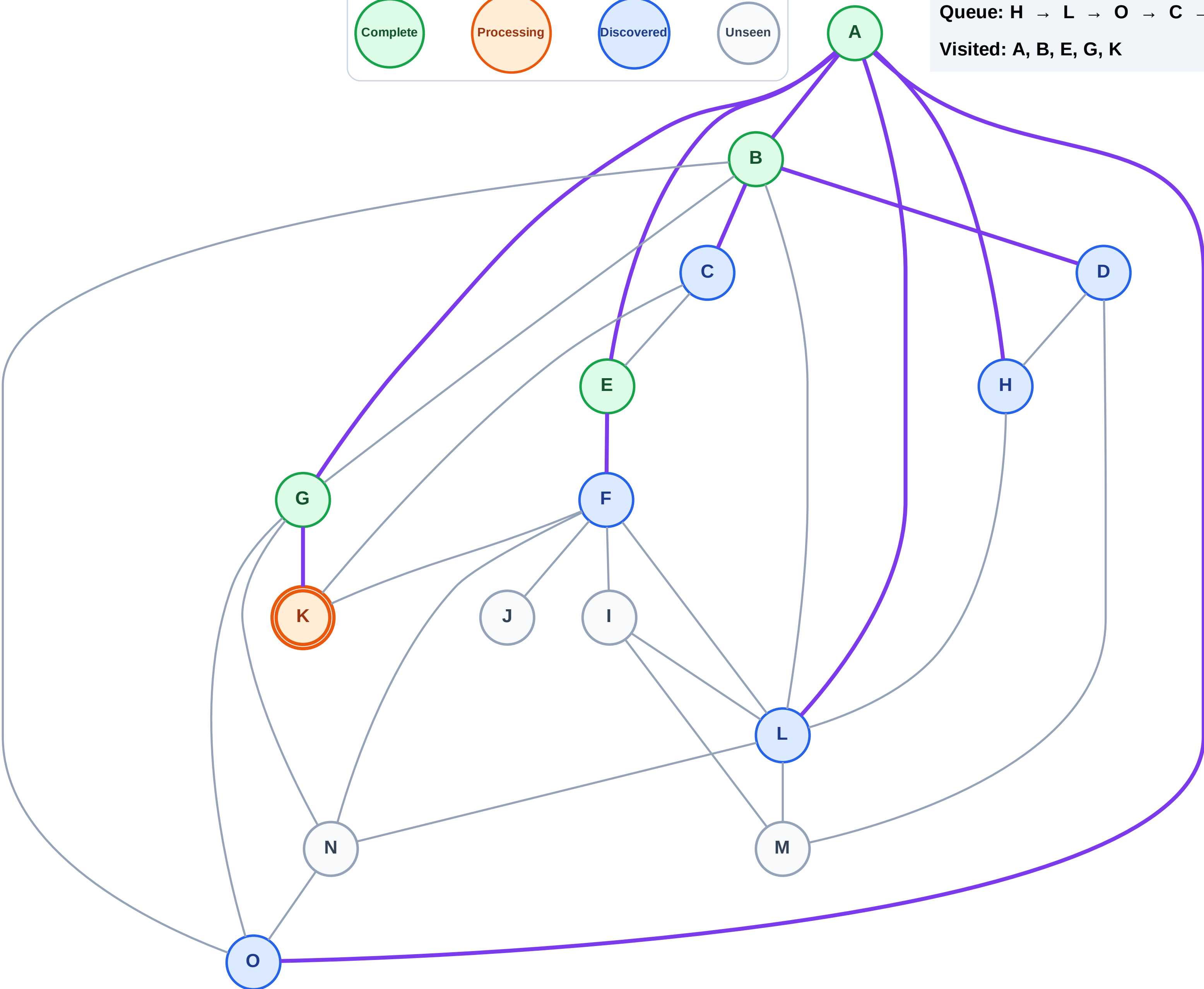
Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → G → K

Legend



Queue: H → L → O → C → D → F

Visited: A, B, E, G, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

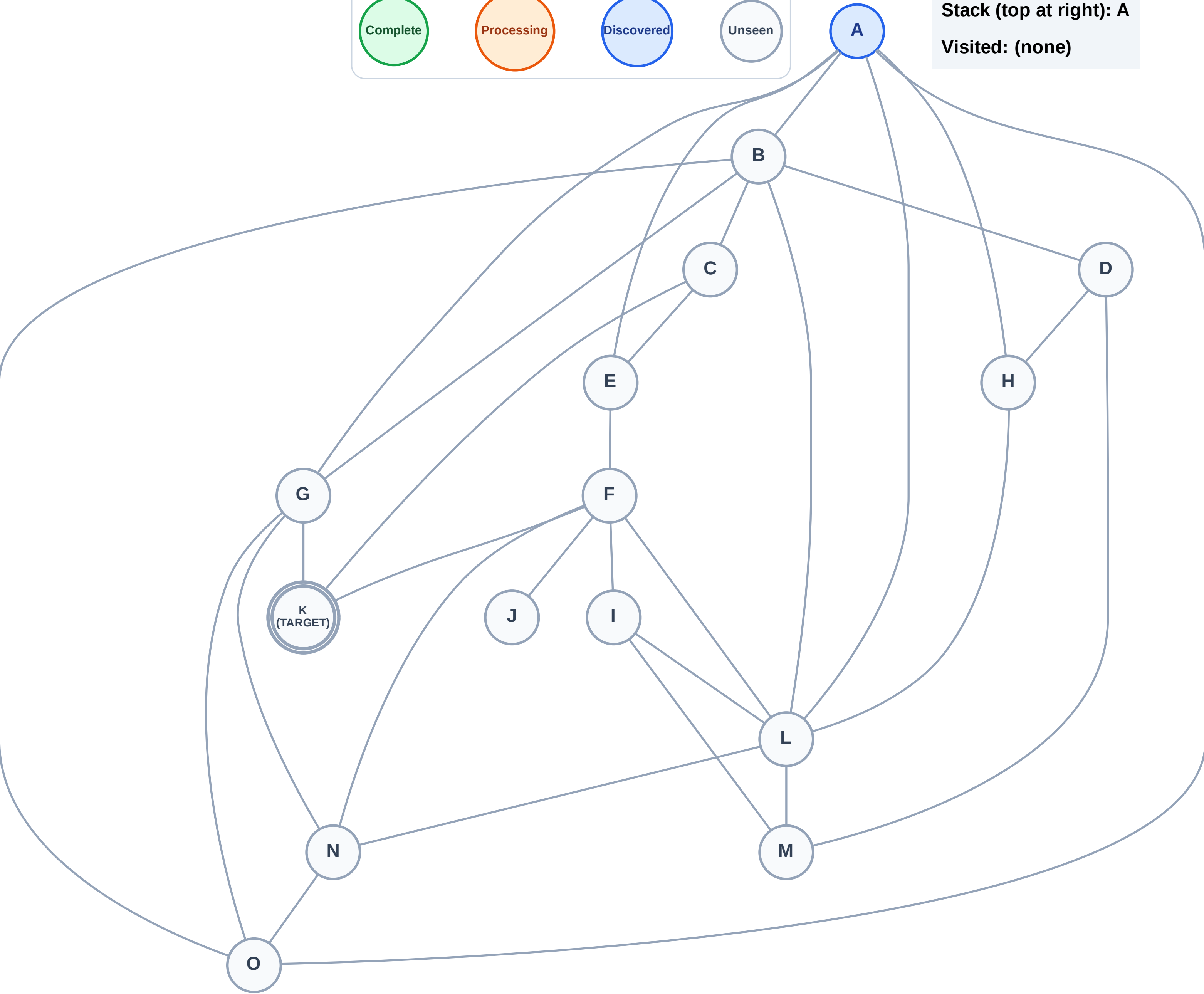
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

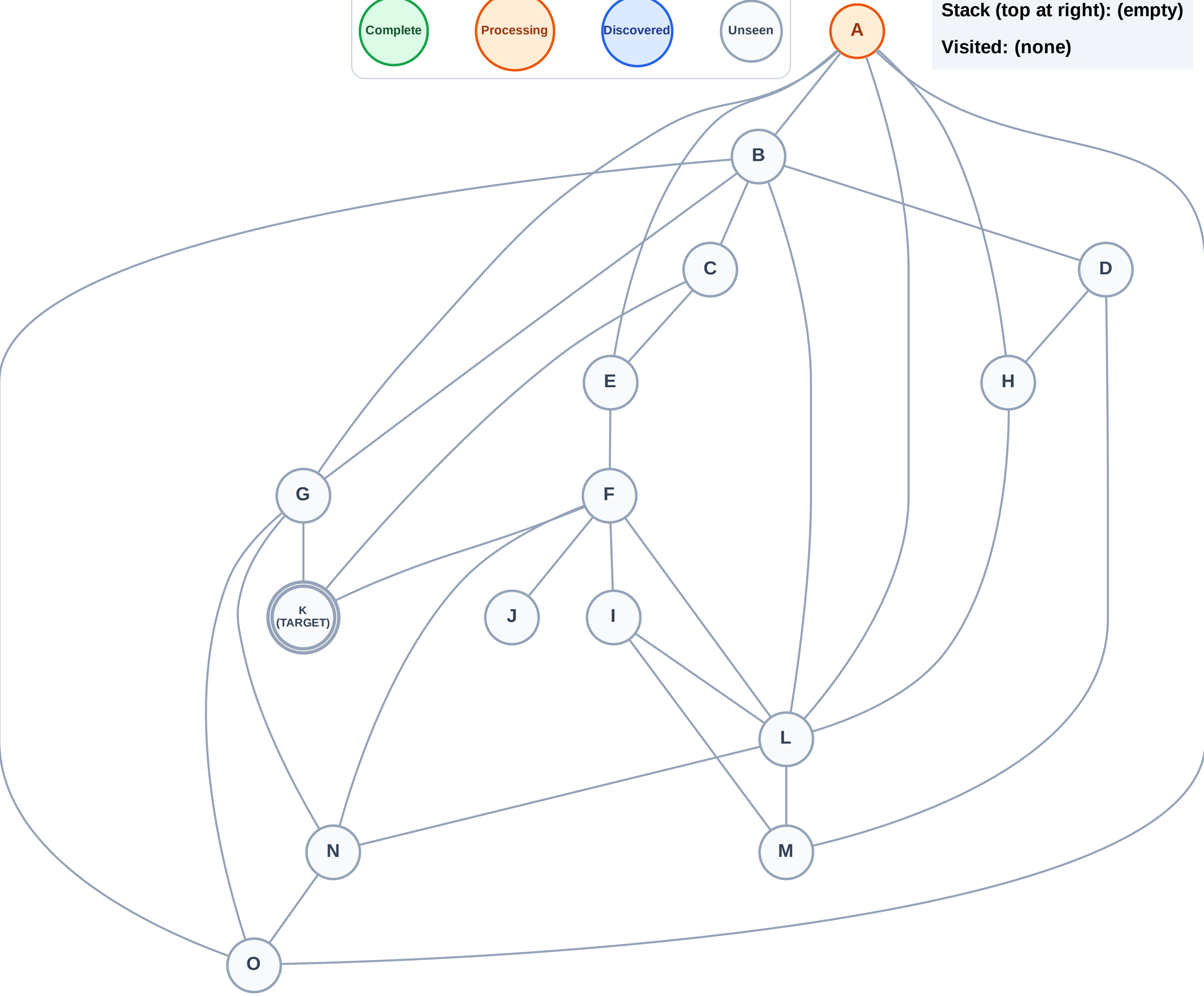
Complete

Processing

Discovered

Unseen

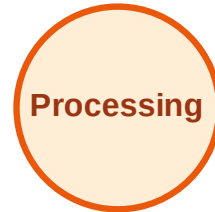
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

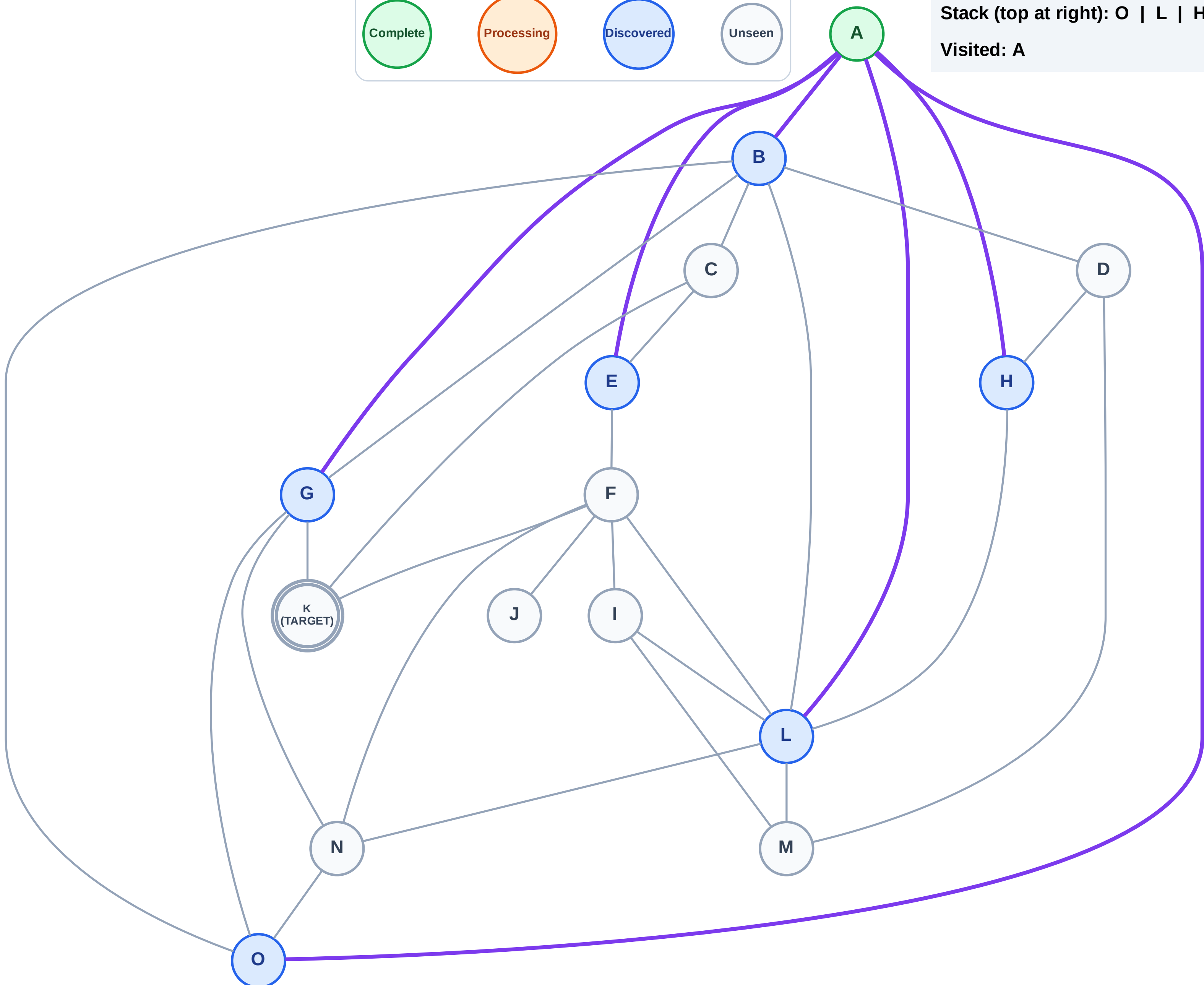
Step 3: Discover O, L, H, G, E, B from A

Legend



Stack (top at right): O | L | H | G | E | B

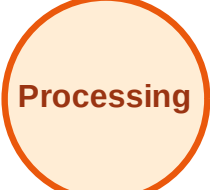
Visited: A



DFS Traversal

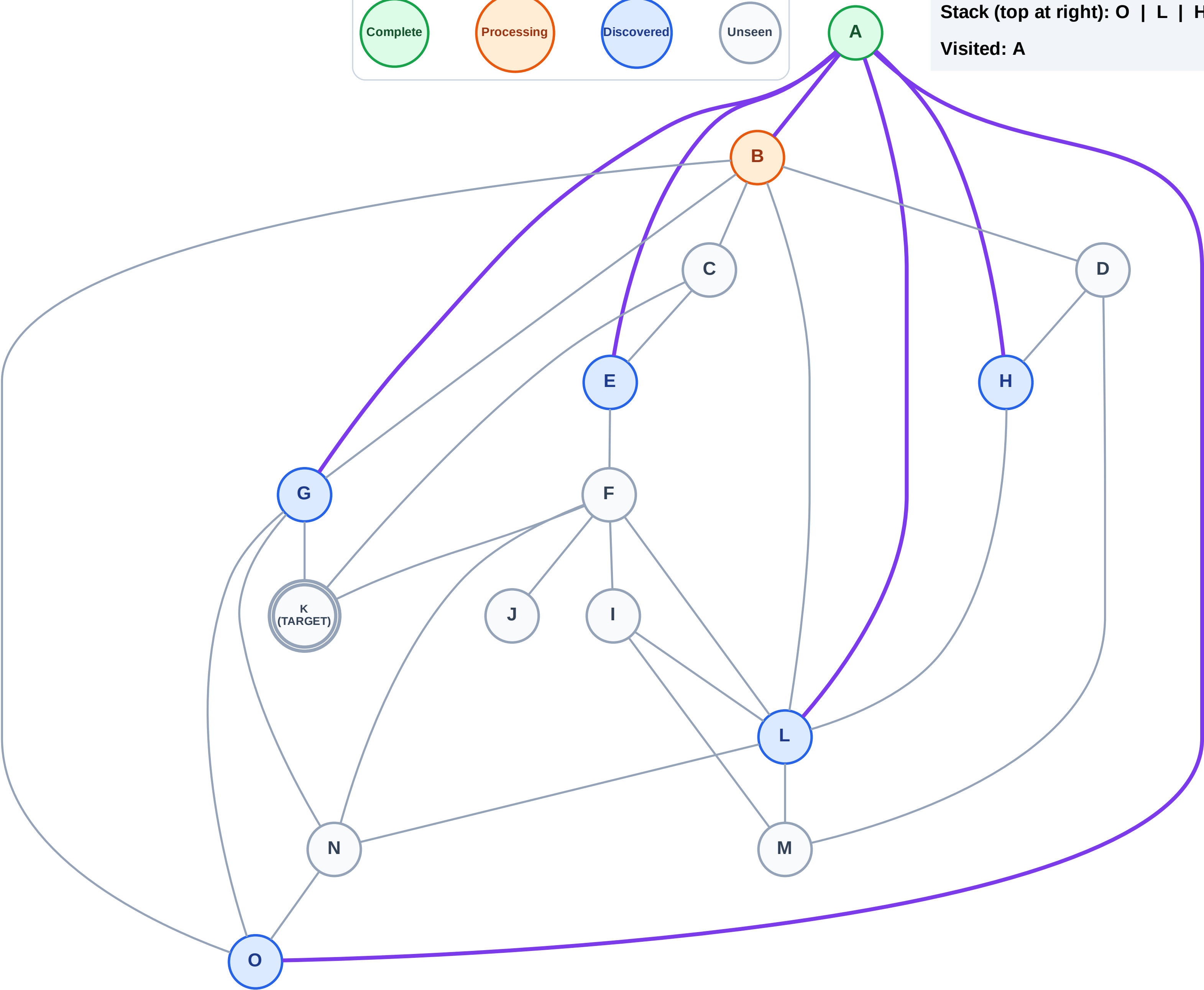
Step 4: Process node B

Legend



Stack (top at right): O | L | H | G | E

Visited: A



DFS Traversal

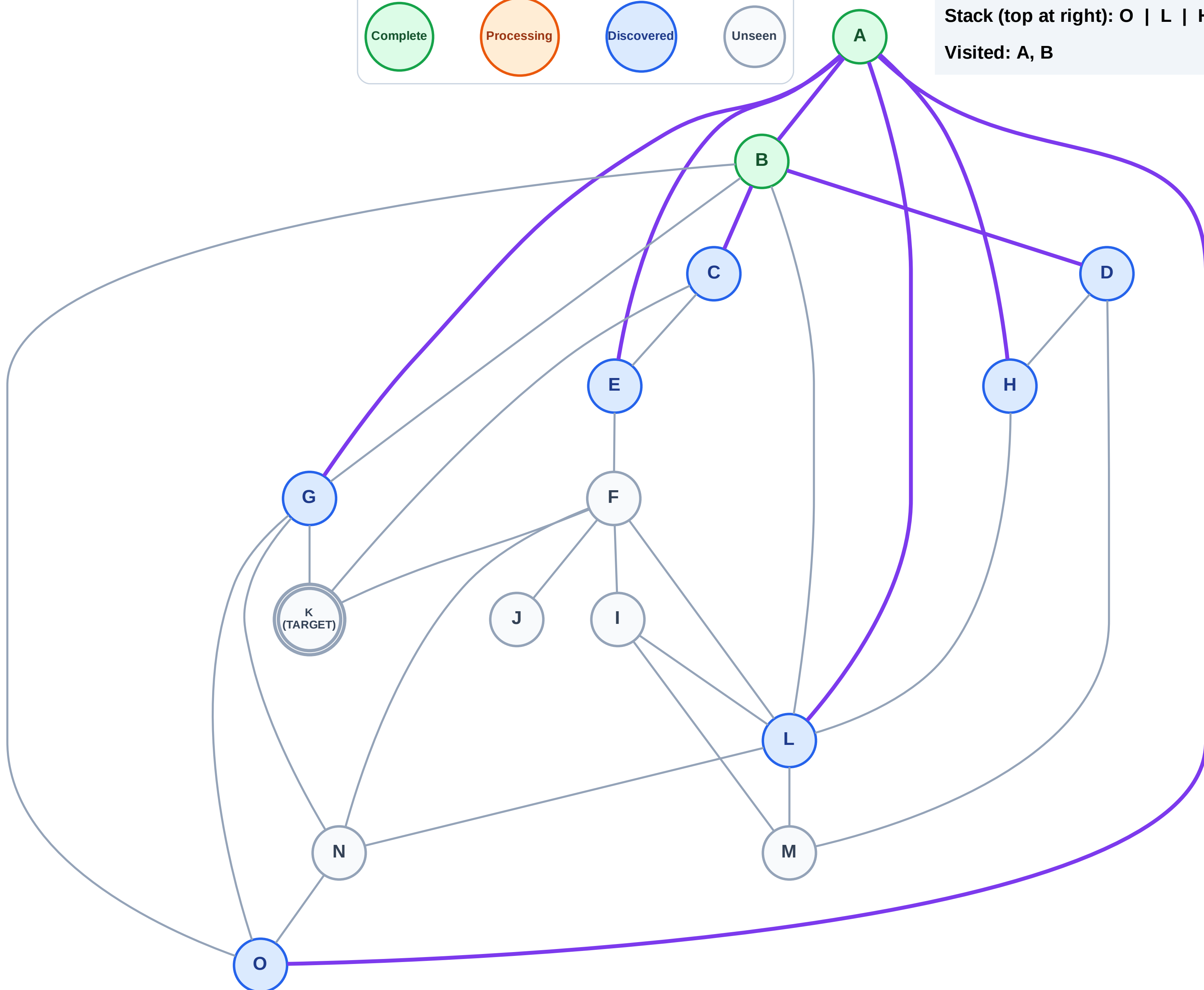
Step 5: Discover D, C from B

Legend



Stack (top at right): O | L | H | G | E | D | C

Visited: A, B



DFS Traversal

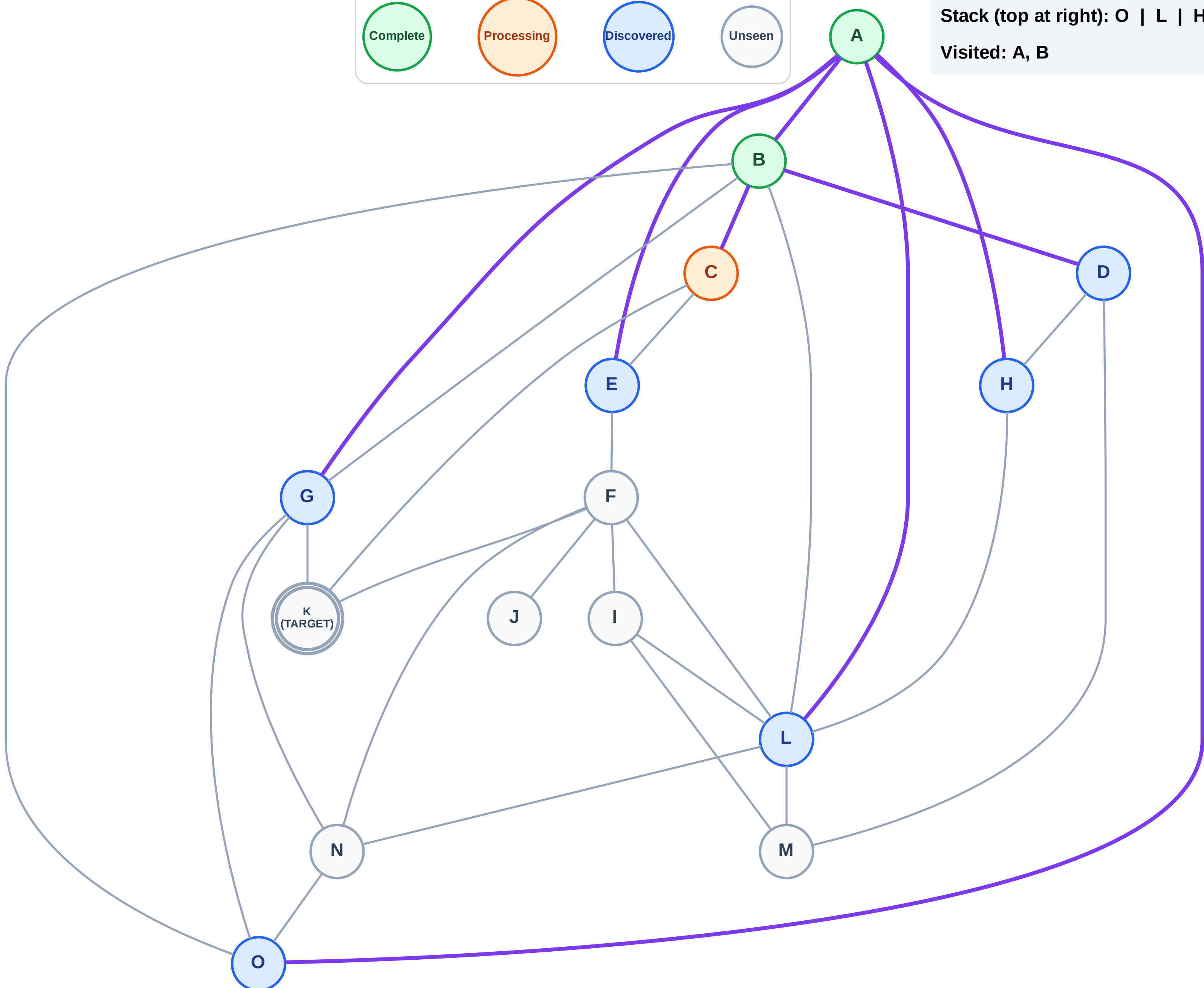
Step 6: Process node C

Legend



Stack (top at right): O | L | H | G | E | D

Visited: A, B



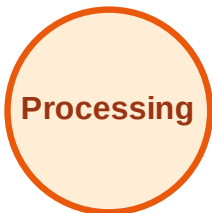
DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): O | L | H | G | E | D

Visited: A, B, C, K

